Q1: 24 Feb (Shift 1) - Single Correct

The population $P=P(t)$ at time $'t'$ of a certain species follows the differential equation $\frac{dP}{dt}=0.5\mathrm{P}-450$. If $\mathrm{P}(0)=850$, then the time at which population becomes zero is :

(1) $\frac{1}{2}\log_e 18$

(2) $2\log_e 18$

(3) $\log_e 9$

(4) $\log_e 18$

Q2: 24 Feb (Shift 2) - Single Correct

Let f be a twice differentiable function defined on R such that $f(0)=1, f'(0)=2$ and $f'(x)\neq 0$ for all $\mathrm{X}\in\mathrm{R}$. If $\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix}=0$, for all $\mathrm{x}\in\mathrm{R}$ then the value of $f(1)$ lies in the interval:

(1) (9,12)

(2) (6,9)

(3) (3,6)

(4) (0,3)

Q3: 24 Feb (Shift 2) - Single Correct

If a curve $y=f(x)$ passes through the point (1,2) and satisfies $x\frac{\mathrm{d}y}{\mathrm{d}x}+y=\mathrm{b}x^4$, then for what value of b, $\int_1^2 f(x)\mathrm{d}x=\frac{62}{5}$?

(1) 5

(2) $\frac{62}{5}$

(3) $\frac{31}{5}$

(4) 10

Q4: 25 Feb (Shift 1) - Single Correct

If a curve passes through the origin and the slope of the tangent to it at any point (x, y) is $\frac{x^2-4x+y+8}{x-2}$, then this curve also passes through the point :

(1) (4,5)

(2) (5,4)

(3) (4,4)

(4) (5,5)

Q5: 25 Feb (Shift 2) - Numerical

If the curve $y=y(x)$ represented by the solution of the differential equation $\left(2xy^2-y\right)dx+xdx=0$, passes through the intersection of the lines, $2x-3y=1$ and $3x+2y=8$, then $|y(1)|$ is equal to

Q6: 26 Feb (Shift 1) - Single Correct

The rate of growth of bacteria in a culture is proportional to the number of bacteria present and the bacteria count is 1000 at initial time $\mathrm{t}=0$. The number of bacteria is increased by 20% in 2 hours. If the population of bacteria is 2000 after $\frac{\mathrm{k}}{\log_e\left(\frac{6}{5}\right)}$ hours, then $\left(\frac{\mathrm{k}}{\log_e 2}\right)^2$ is equal to

(1) 4

(2) 2

(3) 16

(4) 8

Q7: 26 Feb (Shift 1) - Numerical

If $y=y(x)$ is the solution of the equation $e^{\sin y}\cos y\,\frac{dy}{dx}+e^{\sin y}\cos x=\cos x, y(0)=0$; then $1+y\left(\frac{\pi}{6}\right)+\frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right)+\frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right)$ is equal to

Q8: 26 Feb (Shift 1) - Numerical

The difference between degree and order of differential equation that represents the family of curves given by $y^2=a\left(x+\frac{\sqrt{a}}{2}\right), a>0$ is

Q9: 26 Feb (Shift 2) - Single Correct

Let slope of the tangent line to a curve at any point $\mathrm{P}(\mathrm{x}, \mathrm{y})$ be given by $\frac{xy^2+y}{x}$. If the curve intersects the line $x+2y=4$ at $x=-2$, then the value of y, for which the point $(3, y)$ lies on the curve, is :

(1) $-\frac{18}{11}$

(2) $-\frac{18}{19}$

(3) $-\frac{4}{3}$

(4) $\frac{18}{35}$

Answer Key

Q1 (2)	**Q2** (2)	**Q3** (4)	**Q4** (4)
Q5 (1)	**Q6** (1)	**Q7** (1)	**Q8** (2)
Q9 (2)			

Q1 (2)

$\frac{dp}{dt} = \frac{p-900}{2}$

$\int_{850}^{0} \frac{dp}{p-900} = \int_0^t \frac{dt}{2}$

$\ell n \mid P - 900 \|_{850}^{0} = \frac{t}{2}$

$\ell n|900| - \ell n|50| = \frac{t}{2}$

$\frac{t}{2} = \ell n|18|$

$\Rightarrow t = 2 \ln 18$

Q2 (2)

Given $f(x)f''(X) - (f'(x))^2 = 0$

Let $h(x) = \frac{f(x)}{f'(x)}$

$\Rightarrow h'(x) = 0 \quad \Rightarrow h(x) = k$

$\Rightarrow \frac{f(x)}{f'(x)} = k \quad \Rightarrow f(x) = kf'(x)$

$\Rightarrow f(0) = kf'(0) \quad \Rightarrow 1 = k(2) \Rightarrow k = \frac{1}{2}$

Now $f(x) = \frac{1}{2}f'(x) \Rightarrow \int 2dx = \int \frac{f'(x)}{f(x)}dx$

$\Rightarrow 2x = \ln|f(x)| + C$

As $f(0) = 1 \Rightarrow C = 0$

$\Rightarrow 2x = \ln|f(X)| \Rightarrow f(x) = \pm e^{2x}$

As $f(0) = 1 \Rightarrow f(x) = e^{2x} \Rightarrow f(1) = e^2$

Q3 (4)

$\frac{dy}{dx} + \frac{y}{x} = bx^3 \cdot \text{I.F.} = e^{\int \frac{dx}{x}} = x$

$\therefore yx = \int bx^4 dx = \frac{bx^5}{5} + c$

Passes through $(1, 2)$, we get

$2 = \frac{b}{5} + C \ldots\ldots (i)$

Also, $\int_1^2 \left(\frac{bx^4}{5} + \frac{c}{x}\right) dx = \frac{62}{5}$

$\Rightarrow \frac{b}{25} \times 32 + C\ln 2 - \frac{b}{25} = \frac{62}{5} \Rightarrow C = 0 \,\&\, b = 10$

Q4 (4)

$\frac{dy}{dx} = \frac{(x-2)^2+y+4}{(x-2)} = (x-2) + \frac{y+4}{(x-2)}$

Let $x - 2 = t \Rightarrow dx = dt$

and $y + 4 = u \Rightarrow dy = du$

$\frac{dy}{dx} = \frac{du}{dt}$

$\frac{du}{dt} = t + \frac{u}{t} \Rightarrow \frac{du}{dt} - \frac{u}{t} = t$

$I.F = e^{\int \frac{-1}{t} dt} = e^{-Int} = \frac{1}{t}$

$u.\frac{1}{t} = \int t \cdot \frac{1}{t} dt \Rightarrow \frac{u}{t} = t + c$

$\frac{y+4}{x-2} = (x-2) + c$

Passing through (0,0) c = 0

$\Rightarrow (y+4) = (x-2)^2$

Q5 (1)

Given,

$(2xy^2 - y)\,dx + xdx = 0$

$\Rightarrow \frac{dy}{dx} + 2y^2 - \frac{y}{x} = 0$

$\Rightarrow -\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y}\left(\frac{1}{x}\right) = 2$

$\frac{1}{y} = z$

$-\frac{1}{y^2}\frac{dy}{dx} = \frac{dz}{dx}$

$\Rightarrow \quad \frac{dz}{dx} + z\left(\frac{1}{x}\right) = 2$

$\Rightarrow \quad F_x = e^{\int \frac{1}{x} dx} = x$

$\Rightarrow z(x) = \int 2(x)dx = x^2 + c$

$\Rightarrow \frac{x}{y} = x^2 + c$

As it passes through P(2, 1)

[Point of intersection of $2x - 3y = 1$ and $3x + 2y = 8$]

$\therefore \frac{2}{1} = 4 + c$

$\Rightarrow c = -2$

$\Rightarrow \frac{x}{y} = x^2 - 2$

Put $x = 1$

$\frac{1}{y} = 1 - 2 = -1$

$\Rightarrow y(1) = -1$

$\Rightarrow |y(1)| = 1$

Q6 (1)

$\frac{dx}{dt} \propto x$

$\frac{dx}{dt} = \lambda x$

$\int_{1000}^{x} \frac{dx}{x} = \int_0^t \lambda dt$

$\ln x - \ln 1000 = \lambda t$

$\ln(\frac{x}{1000}) = \lambda t$

Put $t = 2, x = 1200$

$\ln(\frac{12}{10}) = 2\lambda \Rightarrow \lambda = \frac{1}{2}\ln\frac{6}{5}$

Now $\ln(\frac{x}{1000}) = \frac{t}{2}\ln(\frac{6}{5})$

$x = 1000e^{\frac{t}{2}\ln(\frac{6}{5})}$

$x = 2000$ at $t = \frac{k}{\ln(\frac{6}{5})}$

$\Rightarrow \quad 2000 = 1000e^{\frac{k}{2\ln(6/5)}} \times \ln(6/5)$

$\Rightarrow 2 = e^{k/2}$

$\Rightarrow \quad \ln 2 = \frac{k}{2}$

$\Rightarrow \frac{k}{\ln 2} = 2$

$\Rightarrow (\frac{k}{\ln 2})^2 = 4$

Q7 (1)

$e^{\sin y}\cos y\frac{dy}{dx} + e^{\sin y}\cos x = \cos x$

Put $e^{\sin y} = t$

$e^{\sin y} \times \cos y\frac{dy}{dx} = \frac{dt}{dx}$

$\Rightarrow \frac{dt}{dx} + t\cos x = \cos x$

$I.F. = e^{\int \cos x dx} = e^{\sin x}$

Solution of diff equation:

$t \cdot e^{\sin x} = \int e^{\sin x} \cdot \cos x dx$

$e^{\sin y} \cdot e^{\sin x} = e^{\sin x} + c$

at $x = 0, y = 0$

$1 = 1 + c \quad \Rightarrow c = 0$

$e^{\sin x + \sin y} = e^{\sin x}$

$\sin x + \sin y = \sin x$

$y = 0$

$\Rightarrow y\left(\frac{\pi}{6}\right) = 0, \quad y\left(\frac{\pi}{3}\right) = 0, \quad y\left(\frac{\pi}{4}\right) = 0$

$\Rightarrow 1 + 0 + 0 + 0 = 1$

Q8 (2)

$y^2 = a\left(x + \frac{\sqrt{a}}{2}\right)$

$2yy' = a$

$y^2 = 2yy'\left(x + \frac{\sqrt{2yy'}}{2}\right)$

$y = 2y'\left(x + \frac{\sqrt{yy'}}{\sqrt{2}}\right)$

$y - 2xy' = \sqrt{2}y'\sqrt{yy'}$

$\left(y - 2x\frac{dy}{dx}\right)^2 = 2y\left(\frac{dy}{dx}\right)^3$

$D = 3$ & $O = 1$

$D - O = 3 - 1 = 2$

Q9 (2)

$\frac{dy}{dx} = \frac{xy^2+y}{x}$

$\Rightarrow \frac{xdy-ydx}{y^2} = xdx$

$\Rightarrow -d\left(\frac{x}{y}\right) = d\left(\frac{x^2}{2}\right)$

$\Rightarrow \frac{-x}{y} = \frac{x^2}{2} + C$

Curve intersect the line $x + 2y = 4$ at $x = -2$ So, $-2 + 2y = 4 \Rightarrow y = 3$

So the curve passes through (-2,3) $\Rightarrow \frac{2}{3} = 2 + C$

$\Rightarrow C = \frac{-4}{3}$

$\therefore$ curve is $\frac{-x}{y} = \frac{x^2}{2} - \frac{4}{3}$

It also passes through $(3, y)$ $\frac{-3}{y} = \frac{9}{2} - \frac{4}{3}$

$\Rightarrow \frac{-3}{y} = \frac{19}{6}$

$\Rightarrow y = -\frac{18}{19}$